

Namma Space: Engineering Research Report — Android-Only Indoor Digital-Twin & Navigation Pipeline (Techfest IIT Bombay 2026-27)

TL;DR

- **Build a two-track pipeline:** (1) an ML frame-selector on-device feeding an **offline π^3 /VGGT $\rightarrow$ GLOMAP-refined $\rightarrow$ 3D Gaussian Splatting** reconstruction rendered in the browser via **PlayCanvas/SuperSplat with SOG compression**; (2) a **mesh/navmesh track** (2DGS $\rightarrow$ recast-navigation-js) for obstacle-aware pathfinding. This is fully achievable Android-only and finishes overnight (in fact 1–2 h) on a single consumer GPU costing well under \$1.
- **The digital twin + web viewer + POI search + walkable pathfinding is a strong, deliverable core.** On-site AR navigation is the hardest part and the biggest risk; ship the cheapest credible version — **sparse visual relocalization (ACE, which maps in 5 min to a 4 MB network at ~97% @5cm/5° on 7-Scenes) at decision points + ARCore VIO between them** — and treat full continuous AR nav as a one-aisle stretch goal.
- **EchoVoid transfers heavily:** ARCore session/pose/DEPTH16 capture, Cloud Anchors, Wi-Fi/magnetic room identity, the ORB+PnP relocalizer (VisualMap.kt), MediaPipe+Gemini detection, and the OpenGL renderer are all directly reusable for frame selection, metric scale, POI labeling, and relocalization. Rebuild only the reconstruction and web-delivery layers.

Key Findings

1. Frame/keyframe selection from handheld video

Smart frame selection materially improves reconstruction versus uniform sampling, and the SOTA is a fusion of blur detection, IMU gating, and view-overlap criteria.

- **Blur/quality:** Laplacian-variance is the classic baseline; successors use FFT high-frequency energy and Lucas-Kanade optical flow to detect motion blur. The **IVS3D** open framework (arXiv:2110.11810) provides intelligent video sampling for 3D reconstruction with pluggable quality metrics, and explicitly notes that naive frame-rate downsampling "will not always result in a good selection of keyframes."
- **IMU-gated selection:** A random-forest classifier trained on inertial data can drop redundant/blurred frames in real time (ResearchGate: "Selection of Key Frames for 3D Reconstruction in Real Time"), though the authors candidly report "unsatisfied performance with respect to ideal frames" — treat IMU gating as a coarse filter, not the sole criterion.
- **Geometric criteria:** Robust keyframe extraction combining frame-to-frame correspondence count, GRIC scores for homography/fundamental matrix, and point-to-epipolar-line cost (Ahmed & Dailey) avoids degeneracies from low-parallax/pure-rotation motion — the common indoor failure.
- **Learned/adaptive (2025):** "Adaptive Keyframe Selection for Scalable 3D Scene Reconstruction in Dynamic Environments" (arXiv:2510.23928) combines photometric + SSIM error with momentum-based dynamic thresholds and shows consistent gains layered on Spann3R and CUT3R — evidence that adaptive selection beats static strategies on Chamfer distance and normal consistency.
- **ARCore signals:** ARCore's own `TrackingState`, tracking-failure reasons (insufficient features, excessive motion, insufficient light), and per-frame VIO motion are free selection inputs — reuse EchoVoid's session plumbing to tag each frame with pose delta and tracking quality.

Verdict: Implement a lightweight on-device filter = (Laplacian/FFT blur score) $\times$ (ARCore tracking-quality gate) $\times$ (baseline/overlap gate from pose deltas), targeting 100–300 frames. This is well-supported and directly reuses EchoVoid's ARCore frame+pose pipeline. **Benchmark it against uniform sampling — the quantified improvement is a graded, citable result.**

2. Video/images $\rightarrow$ 3D reconstruction — the current landscape

Classical SfM/MVS: COLMAP (incremental, most robust/accurate, but the slowest). **GLOMAP** (ECCV 2024, colmap/glomap, BSD) achieves "approximately 8% higher recall and scores 9 and 8 additional points in AUC at

the 0.1m and 0.5m thresholds, respectively, compared to COLMAP, which is also one order of magnitude slower" ([arXiv](#)) (Pan et al., arXiv:2407.20219); on ETH3D SLAM it runs in ~133.5 s and reconstructs scenes COLMAP fails on. Both still struggle on textureless indoor walls and low-parallax motion. **MP-SfM** (arXiv:2504.20040) adds monocular surface priors specifically for the low-parallax/textureless indoor case; MAST3R-SfM handles low overlap.

Feed-forward geometry foundation models — the game-changer for this project:

- **VGGT** (CVPR 2025 Best Paper, facebookresearch/vggt): "It accepts up to hundreds of images and predicts cameras, point maps, depth maps, and point tracks for all images at once in less than a second" ([GitHub](#)) ([TheCVF](#)) — a feed-forward regime at **~0.2 seconds per reconstruction**. Weights are non-commercial research only (verify terms).
- **π^3 /Pi3** (ICLR 2026, yyfz/Pi3, arXiv:2507.13347) is permutation-equivariant and needs no reference frame: "On the Sintel benchmark, π^3 reduces the camera pose estimation ATE from VGGT's 0.167 down to 0.074 and improves the scale-aligned video depth absolute relative error from 0.299 to 0.233," ([arXiv](#)) running at **57.4 FPS compared to DUST3R's 1.25 FPS and VGGT's 43.2 FPS**. ([arXiv](#)) On ScanNet ATE is 0.031 vs VGGT's 0.035. Critically, per the repo, "the model weights are restricted to non-commercial research and educational purposes only." ([GitHub](#)) π^3 is initialized from VGGT.
- Ecosystem: Fast3R (thousands of images, no global alignment), CUT3R (streaming), Spann3R, MAST3R/MASt3R-SfM, MoGe/MoGe-2, StreamVGGT, plus 2026 successors (VGGT-Long for km-scale, VGGT-Motion). Frame-count scaling is limited by GPU memory (attention over all tokens); chunk-and-align (VGGT-Long) is the mitigation for large stores.

3D Gaussian Splatting for indoor capture:

- Original **3DGS** (Kerbl 2023) — real-time photorealistic; the base.
- **InstantSplat** (arXiv:2403.20309, NVlabs) fuses DUST3R/MASt3R/VGGT priors with 3DGS for pose-free sparse-view reconstruction, "increasing the SSIM from 0.68 to 0.89 and reducing the ATE from 0.055 to 0.011, while significantly accelerating the optimization from around 2 hours to approximately 1 minute compared to Nope-NeRF"; ([AI Models](#)) in a 3-view Tanks&Temples setting SSIM rose from 0.3755 (COLMAP+3DGS) to 0.7624. ([Liner](#)) It supports 3DGS/2DGS/Mip-Splatting. This is the fastest path from unposed phone frames to a splat.
- **2DGS** (Huang 2024) — geometrically accurate surfels, meshable. **Mip-Splatting** — anti-aliasing. **ScaffoldGS** — anchor-based, better for structured indoor scenes. **Hierarchical 3DGS** (Kerbl 2024; SmallCity ~ 1.8×10^7 Gaussians) ([arXiv](#)) with LoD for very large scenes; OctreeGS/CityGaussianV2 for city scale. **Taming 3DGS** — bounded densification for speed.
- SLAM-integrated: SplatAM, MonoGS, Photo-SLAM (real-time but lower final quality).
- Tooling: **gsplat/nerfstudio** (Apache-2.0 — use this to avoid the non-commercial weights problem in the *optimizer*), **Brush** (Rust/WebGPU, cross-platform), **Postshot** (Windows).

Mesh/TSDF alternatives (when a mesh beats a splat — collision, navmesh, occlusion, small files): NeuralRecon, ARCore Scene Semantics, RTAB-Map, Open3D TSDF, SuGaR (splat→mesh), 2DGS meshing. **You need both:** the splat for the photorealistic viewer and a mesh for navmesh/collision.

Consumer apps — what actually works Android-only, no LiDAR:

- **Scaniverse** (Niantic, free) — on-device Gaussian splatting AND photogrammetry on Android; splat/photogrammetry modes work on non-LiDAR phones (some older devices can't create splats but can view them); free classic mode with mesh export and no subscription; web/AR viewer.
- **Polycam** — photogrammetry + splats, iOS and Android, cloud meshing; free tier exports GLTF only (OBJ/FBX/STL behind paywall).
- **RealityScan** (Epic, free, no paid mobile tier) — photogrammetry, strong on Android.
- **KIRI Engine** — Android-first, photogrammetry + splats.

- **Luma AI** — NeRF/splats but largely iOS.
- **Matterport on phone** — Cortex AI infers depth from 2D/360; ~1% accuracy with Pro2, 4-8% with 360 cameras.

VERDICT for a ~200-2000 m² retail/office space, Android-only, overnight on one consumer/cloud GPU:

Capture handheld video → on-device frame selection (100-300 frames) → π^3 (or VGGT) **feed-forward init** (poses + dense point cloud in ~0.2-1 s) → **GLOMAP refinement with ARCore poses injected** for metric scale/robustness → **3DGS optimization via the InstantSplat/gsplat pipeline** → **2DGS pass for the mesh**. Deliver the splat compressed as **SOG** for the web. For spaces beyond a single room, tile with Hierarchical-3DGS LoD. Flag the **non-commercial license** on VGGT/ π^3 *weights* (fine for a hackathon; use Apache-2.0 gsplat for the optimizer).

3. Metric scale and drift

Monocular reconstruction is scale-ambiguous; several Android-native levers inject true metric scale and a level floor:

- **ARCore pose priors as SfM constraints** — ARCore VIO poses are metric. Inject them into COLMAP/GLOMAP as pose priors (COLMAP supports pose priors / rig constraints) to fix scale and improve low-parallax robustness. This is your single most valuable lever and reuses EchoVoid's verified pose stream.
- **ARCore Depth API** (depth-from-motion, works without ToF) — metric depth per frame; EchoVoid already decodes DEPTH16 at 180/180 frames. Use it to scale the point cloud and validate. (Note: ARCore's *Geospatial* Depth reaching 65 m is outdoor-only and irrelevant indoors.)
- **Gravity alignment:** accelerometer gravity vector → level the floor; fit a floor plane (RANSAC in Open3D) for level ground.
- **Known-object scale references and loop closure** for drift correction.
- Injecting known poses into 3DGS: InstantSplat and pose-conditioned Pi3X accept camera priors.

4. Rendering in the browser

- **PlayCanvas Engine + SuperSplat** (MIT) is the strongest "fast + web-deliverable" option: browser-based splat editing, cleanup, publishing, and the **SOG (Spatially Ordered Gaussians)** format. Per PlayCanvas, a skate-park scan "has 4 million Gaussians but is compressed to a mere 42MB with SOG. The original PLY was 1GB, meaning a ~95% reduction in file size"; SOG "stores splat data in Morton order meaning it's 'GPU-ready'" and "only needs WebGPU so it runs anywhere." Streamed SOG provides LOD/progressive loading (SuperSplat auto-selects the most detailed level under 20M splats). The `splat-transform` CLI reads/writes PLY, SOG, SPZ, KSPLAT, LCC, and generates collision meshes and LOD.
- Alternatives: **antimatter15/splat** and **gsplat.js** (simple WebGL), **Spark** (World Labs), **Babylon.js** GS support, **three.js** loaders + **luma-web**, **Niantic 8th Wall** for WebAR, **Potree** for point clouds. Mesh compression: Draco/meshopt/KTX2.
- **First-person navigation:** WASD/touch controller in PlayCanvas or three.js; collision against the extracted **mesh** (not the splat) using a simplified collision mesh from `splat-transform`; render the path as an on-floor polyline. Keep splat rendering and navmesh queries separate — pathfinding runs on the lightweight navmesh while the splat renders.
- Realistic performance: SuperSplat 3.0 uses WebGPU with >10× less idle memory; mid-range laptops handle a few million Gaussians; phones need LOD/Streamed SOG and Gaussian budgets.

5. Walkable space extraction & pathfinding

- **Free-space extraction:** project the mesh/point cloud to a 2D occupancy grid with height slicing (keep points ~10 cm-2 m to find obstacles; floor = lowest plane). Fit the floor via RANSAC (Open3D/CloudCompare).
- **Floorplan/layout:** **RoomFormer** (CVPR 2023, ywyue/RoomFormer) does single-stage floorplan reconstruction from point-cloud density maps, `Emergent Mind` outputs room polygons + doors/windows, `arXiv` trained on **Structured3D** and SceneCAD, `arXiv` and is faster than HEAT/MonteFloor. `arXiv` Alternatives: HEAT, FRI-Net, PolyRoom, PolyDiffuse.

- **Navmesh: recast-navigation-js** (WASM port of Recast/Detour, isaac-mason, v0.43+) generates navmeshes from three.js meshes at runtime or offline; supports agent radius (clearance/inflation for shoulders/trolleys), off-mesh links (doorways), tiled navmeshes (multi-floor), temporary obstacles, and crowd simulation. **three-pathfinding** (donmccurdy, PatrolJS-based) does A* over an imported navmesh. Query Detour for A*/funnel-smoothed (any-angle) paths.
- **Clearance/doorways:** set Recast agent radius to ~0.3–0.4 m (person) or larger (trolley); doorways handled by navmesh connectivity + off-mesh links.

6. POIs, spatial indexing & semantic search

Two paths, use both:

- **Pragmatic authoring (ship this first):** a store-owner tool to click in the 3D view and drop/label POIs (raycast into the mesh → 3D coordinate → named POI). Store POIs with 3D position + label + synonyms. Text search with fuzzy matching + a synonym map ("oats" → breakfast aisle). Straightforward full-stack build that plays to the team's strengths.
- **Open-vocabulary 3D understanding (differentiator):** **LangSplat** (CVPR 2024) builds a 3D language field on Gaussians with SAM hierarchical semantics, far faster than LERF (reported hundreds-fold speedup at 1440×1080). **Feature-3DGS**, **Gaussian Grouping**, **OpenGaussian**, **LEGaussians** are alternatives. **ConceptGraphs** (ICRA 2024) builds open-vocabulary 3D scene graphs from posed RGB-D + 2D foundation models (**Emergent Mind**) (SAM+CLIP+LLM) — ideal for object-centric POIs and relational queries; HOV-SG adds hierarchy with a 75% size reduction vs dense maps. (**Semantic Scholar**) For 2D detection lifted to 3D: **Grounding-DINO**, **OWLv2**, **SAM/SAM2**. EchoVoid's MediaPipe EfficientDet-Lite0 + Gemini open-vocabulary boxes transfer directly.
- **Retail-specific:** shelf/planogram recognition + product localization; use Grounding-DINO/OWLv2 for shelf-face and product detection, project into the 3D twin.

7. On-site AR navigation on Android — the hardest part (honest assessment)

Visual relocalization against a prebuilt map:

- **ACE** (CVPR 2023, nianticlabs/ace) — scene-coordinate regression: "ACE compiles the scene into 4MB worth of MLP weights during 5 minutes on a single consumer-grade GPU," (**Nianticlabs**) and is "the only approach that 1) achieves top accuracy 2) in less than 10 minutes per scene 3) with less than 10MB per scene." (**TheCVF**) It hits ~97% of frames within 5 cm/5° on 7-Scenes with a 9-layer, 512-channel MLP as the map. Successors: ACE-Zero, GLACE, Marepo, ACE-G (better generalization). **Best accuracy/size/speed tradeoff for a retail-scale map.**
- **hloc** (SuperPoint + SuperGlue/LightGlue + NetVLAD) — ~95.7% @5cm/5° on 7-Scenes but a ~3 GB map, heavier query (retrieval + matching + PnP-RANSAC), sub-real-time on mobile. Best run as a thin-client to a server.
- Splat-based localization: SplatLoc, GSLoc, 6DGS, iComMa — research-grade, promising but immature.
- **LaMAR** (ECCV 2022) shows real-world large-scale AR relocalization is far from saturated and benefits from radio (Wi-Fi/BT) + sequence fusion — directly relevant to a changing store.

ARCore options: Cloud Anchors (keyless; EchoVoid already uses; measured proxy position error ~0.16 m, resolve latency sub-second to ~1 s, needs good lighting; up to 40 concurrent ops but EchoVoid found a 3-anchor practical limit). Geospatial API/VPS/Streetscape Geometry are **outdoor** (Street View coverage; Streetscape Geometry is building/terrain within 100 m). Google notes VPS "coverage is important in areas with low GPS accuracy like indoors," but there is no supported *store-interior* VPS — **do not rely on VPS indoors**. Persistent local anchors + Cloud Anchors are the ARCore-native indoor tools.

Sensor fusion between relocalizations (quantitative, for the honest feasibility case):

- **ARCore VIO drift ~1–3% of distance** in good conditions (a corridor test showed ~0.5 m error over 19.1 m ≈ 2.6%), degrading to ~10–17% in low-texture/rotation-heavy segments (the peer-reviewed *Sensors* 2022

22:9873 benchmark reports ARKit's cleanest figure of ~0.02 m/s drift; ARCore lacked a clean %-of-length number).

- **Wi-Fi RTT/FTM:** Android docs — "If you measure the distance to three or more access points, you can use a multilateration algorithm... The result is typically accurate within 1-2 meters." (Android Developers) Academic WiNar (Elsevier PMC) reports sub-meter: "average localization error of 0.73 m and 0.77 m for the two testbeds." (ScienceDirect) **But it requires 802.11mc-capable APs the store may not have.**
- **PDR:** ~1-5% distance error (TapeLine: 1.43% walking-distance error); ~1-1.3 m absolute over ~150 m with good heading (Kang et al., *Sensors* 2022).
- **Magnetic:** ~1 m (Oriient vendor claim), 1-3 m (IndoorAtlas), 0.6-0.8 m (academic, needs a fingerprint map). EchoVoid already has magnetic-field room identity/maps and Wi-Fi fingerprinting (21/21 across three buildings).

HONEST VERDICT: A 2-4 person student team **can** get reliable on-site AR nav in a store, but only the **cheapest credible version:** sparse visual relocalization (an ACE map, or hloc thin-client) at **decision points** (entrance, aisle ends) to reset pose, + **ARCore VIO** for continuous tracking between them, + arrows/blue-line on the floor via ARCore + SceneView/Filament. EchoVoid's anchor-relative offset relocalization and Wi-Fi/magnetic room identity are the fallback when vision fails. Full continuous VPS-grade AR nav is not realistic in the timeframe — scope it as a one-aisle stretch demo.

8. Existing solutions & competitors

- **Matterport:** SOP with Pro2/Pro3 or 360 cameras; Cortex AI infers depth from 2D/360. Pro3 ~2,000-4,000 sqft/hr, (Scena) ~1% accuracy (20 mm at 10 m); (TAVCO) 360-camera workflow ~7,000-8,000 sqft/hr with a tripod (10,000-20,000 without), (Geo Week News) 4-8% ("8% max error") error. (Geo Week News) Cloud processing hours to 24-48 h. Fails on reflective/glass and needs good lighting.
- **NavVis VLX3** (wearable LiDAR SLAM): 1,000-4,500 m²/hr at 5 mm (Scena) — dedicated hardware, out of scope, but a useful accuracy ceiling.
- **Google Maps Indoor Live View** (VPS from Street View + user imagery), **Apple Indoor Survey/IMDF**, **Pointr / Mappedin / Dent Reality / Oriient / IndoorAtlas / Situm / Meridian / Jibestream** — sensing varies: BLE beacons (Pointr), magnetic (Oriient/IndoorAtlas), VPS/camera (Dent Reality), Wi-Fi/BLE fusion. **Your differentiator: no beacons, no special hardware — a photorealistic twin + semantic search + AR nav from a single phone video.**
- Open-source to study/fork: nerfstudio, gsplat, InstantSplat, SuperSplat, recast-navigation-js, ConceptGraphs, RoomFormer, hloc, ACE.

9. Capture SOP (explicitly graded)

Evidence-based guidance for a non-technical person with an Android phone:

- **Overlap:** interiors need **high overlap — 80-90% frontal** (Pix4D: "Terrestrial images are strongly recommended for interior reconstruction. A high overlap (90%) is needed"). Polycam interior guidance: 30-50%+ minimum, more is better.
- **Pattern:** **triple perimeter + connective passes + orbital passes around key areas** (Pixpro interior case: "Each space must have at least one complete triple perimeter"). For a store: walk each aisle down both sides filming the shelf face on the *opposite* side, plus a perimeter loop, plus loop closures at aisle ends. "Walk on the perimeter of the room and shoot in the opposite direction" for overlap on textureless walls (3Dflow) (3Dflow).
- **Walking speed:** slow, steady; camera at chest/eye height with a slight downward pitch to catch floor + shelf; avoid fast pans (motion blur).
- **Exposure:** lock exposure/AF; uniform lighting; avoid blown-out windows/freezer glass.
- **Reflective/glass/textureless** (freezers, glass doors, white walls): the classic failure mode — capture from multiple angles, add matte/texture references where possible, accept noisy regions.

- **Time:** a supermarket aisle grid realistically takes ~20–45 min of careful filming for a few hundred m²; Matterport's ~2,000–4,000 sqft/hr is a useful benchmark.

10. UX, onboarding & gamification

- **Real-time coverage/quality feedback:** "painting" surfaces as captured, progress rings, ghost trails, ARCore tracking-state prompts ("move slower", "more light", "point at a textured area"). Polycam/Scaniverse/RealityScan/Matterport all use coverage visualization — study and emulate.
- **Gamification:** HCI/citizen-science literature shows points/streaks/completion meters/badges reliably improve *engagement and volume*, with generally positive (but context-dependent) effects on *data quality* when the reward is tied to quality-relevant actions — e.g., a completion meter that rewards coverage directly improves coverage. Keep it plain-language and accessible.
- **Consumer side:** in-store search-to-shelf flow; shopping list → optimal multi-stop route (**TSP over POIs**, solved with nearest-neighbor + 2-opt, or Google OR-Tools); AR "blue line"/arrow wayfinding conventions (Google Live View style).

11. System architecture & cost

- **Android capture app** (Kotlin, ARCore — reuse EchoVoid) → on-device frame selection → **chunked HEVC upload** → **processing queue** (Redis/Celery or a simple job runner) → **GPU worker** → **artifact store** (S3/R2: SOG splat, mesh, navmesh, POI JSON, floorplan) → **web viewer** (PlayCanvas) + **POI/navmesh service** → **phone AR client**.
- **GPU cost:** RTX 4090 rents ~\$0.34/hr on-demand (RunPod Community) up to ~\$0.55/hr (Spheron), and ~\$0.16–0.31/hr spot (Salad/Vast.ai); A100 80GB ~\$1.07–1.39/hr on-demand, ~\$0.60–0.67/hr spot. **For 300 frames:** π^3 /VGGT init is sub-second to seconds; 3DGS optimization on a 4090 is ~20–60 min room-scale; a 2DGS mesh pass adds ~20–30 min. **A full scene costs well under \$1 of GPU and finishes in 1–2 hours** — comfortably overnight, and survivable for Round 3's same-day capture-to-deploy if you keep a warm 4090 provisioned.
- **Free/cheap tiers:** Colab (free/Pro), Lightning, RunPod, Vast.ai. On-device: frame selection, ARCore pose/depth, the AR client.

12. Accuracy evaluation (for judges)

- **Reconstruction:** PSNR/SSIM/LPIPS on held-out views (standard 3DGS metrics); geometric error vs **tape-measure ground truth** (report cm error on known distances — highly persuasive to judges).
- **Tracking:** ATE/RPE.
- **Navigation:** path-length ratio (SPL), success rate; **relocalization recall at (0.25 m, 2°) and (0.5 m, 5°).**
- **Datasets for quick sanity-testing without capturing:** ScanNet/ScanNet++, Replica, Matterport3D, ARKitScenes, 7-Scenes (relocalization — fastest reloc sanity check), Aria Digital Twin, Hilti, LaMAR. 7-Scenes and Replica are fastest for reloc and reconstruction respectively.

Details

Recommended architecture (named components & why chosen over alternatives)

Capture (Android/Kotlin, reuse EchoVoid ARCore stack)

- ARCore session capturing camera image + DEPTH16 + pose + tracking-state per frame (EchoVoid: verified 180/180). *Why over a plain video recorder:* metric poses + depth solve scale and enable pose-prior injection; already built.
- On-device frame selector: Laplacian/FFT blur × ARCore tracking gate × pose-baseline gate → 100–300 frames. *Why over uniform sampling:* IVS3D and adaptive-keyframe literature show measurable reconstruction gains; runs on-device.

Reconstruction (offline GPU worker, RTX 4090)

- π^3 (or VGGT) feed-forward init → poses + dense point cloud in ~0.2-1 s. *Why over COLMAP-from-scratch*: robust on textureless/low-parallax indoor where incremental SfM fails; π^3 has the best pose accuracy (Sintel ATE 0.074 vs VGGT 0.167) and no reference-frame fragility. *Caveat*: non-commercial weights.
- **GLOMAP** refinement with **ARCore pose priors injected** for metric scale. *Why over COLMAP*: 1-2 orders faster, on-par/better accuracy (8% higher recall).
- **gsplat/nerfstudio 3DGS** (Apache-2.0) via an **InstantSplat-style** init for the photorealistic splat; **2DGS** pass for a meshable surface. *Why*: gsplat's permissive license sidesteps the VGGT/ π^3 weight restriction in the optimizer; InstantSplat is the fastest sparse-view path; 2DGS yields the navmesh/collision mesh.
- **SuGaR/2DGS meshing** → collision + navmesh mesh.

Web delivery

- **SuperSplat/PlayCanvas** publish; **SOG** compression + Streamed SOG LOD. *Why over raw PLY or three.js-only*: MIT, ~95% size reduction (1 GB → 42 MB), WebGPU performance, in-browser editing/cleanup.
- **recast-navigation-js** navmesh from the 2DGS mesh; **three-pathfinding/Detour A*** with agent-radius clearance. *Why over hand-rolled grid A**: WASM, runtime or offline, agent radius = trolley clearance, off-mesh links = doorways, tiled = multi-floor.
- **RoomFormer** floorplan for the 2D minimap/room labels.

POI & search

- Store-owner click-to-place authoring tool (raycast to mesh); POI store with synonyms; fuzzy text search; **TSP (nearest-neighbor + 2-opt / OR-Tools)** for shopping-list routing.
- Stretch: **LangSplat** or **ConceptGraphs** for open-vocabulary semantic search; **Grounding-DINO/OWLv2** shelf/product detection (reuse EchoVoid's MediaPipe+Gemini boxes).

AR client (Android, reuse EchoVoid)

- **ACE** scene-coordinate map (4 MB, 5-min mapping) for relocalization at decision points, or **hloc** thin-client; **ARCore VIO** between; **Cloud Anchors + Wi-Fi/magnetic room identity** (EchoVoid) as fallback; arrows/blue-line via ARCore + SceneView/Filament.

Three biggest technical risks (ranked) + mitigations/fallbacks

1. **On-site AR navigation reliability** (highest). VIO drifts 1-3%+ (up to ~17% in bad segments); indoor VPS is unavailable; relocalization in a changing store (moved stock, crowds) is genuinely hard (LaMAR confirms). *Mitigate*: scope to sparse relocalization at decision points + VIO between; use EchoVoid's Wi-Fi/magnetic room identity + Cloud Anchors as coarse fallback. *Fallback*: demo AR nav in one well-mapped aisle; ship web-viewer navigation as the primary deliverable.
2. **Textureless/reflective indoor reconstruction failure** (freezers, white walls, glass). Incremental SfM fails on low-parallax/textureless. *Mitigate*: π^3 /VGGT feed-forward init + ARCore depth/pose priors + MP-SfM surface priors; enforce the capture SOP (perimeter-opposite-direction, 80-90% overlap). *Fallback*: TSDF/mesh-only for problem regions; accept noisy glass.
3. **Scale/drift and web performance at 2000 m²**. Monocular scale ambiguity; millions of Gaussians choke phones. *Mitigate*: ARCore metric pose injection + gravity/floor-plane alignment; Hierarchical-3DGS LoD + Streamed SOG. *Fallback*: tile the store, load per-room.

EchoVoid: reuse vs rebuild

Reuse directly: ARCore session + camera + DEPTH16 + pose capture (frame-selector foundation, metric scale, pose priors); DEPTH16 decoding; pose→room-frame geometry; ARCore Cloud Anchors (keyless, rooms.json) for AR anchoring/fallback; Wi-Fi fingerprint room identity (21/21); magnetic-field room identity/maps; **ORB keyframe + PnP VisualMap.kt — wire it up; it is exactly the sparse visual relocalizer you need at decision points**; OpenGL renderer (point clouds, floor lattices, billboarded thumbnails — reuse for capture-coverage viz and POI thumbnails); MediaPipe EfficientDet-Lite0 + Gemini open-vocabulary detection (POI/product labeling);

Gemini Live voice agent + 15 tool calls (voice-driven search-to-shelf); binaural spatial-audio beacon (accessibility differentiator). **Rebuild:** the reconstruction pipeline (new: π^3 /VGGT + GLOMAP + gsplat 3DGS); the web viewer + SOG delivery (new: PlayCanvas); navmesh/pathfinding (new: recast-navigation-js); POI authoring UI (new web tool). **Address EchoVoid's known limits:** max 3 Cloud Anchors + cloud dependency → make ACE/hloc *local* relocalization primary, Cloud Anchors only coarse fallback; unknown resolve accuracy → measure it (reloc recall at 0.25 m/2°).

Per-round plan (Techfest deadlines)

- **Round 1 (submission 31 Oct 2026):** Working capture app (frame selection on EchoVoid stack) + offline reconstruction (π^3 /VGGT → gsplat 3DGS → SOG) + PlayCanvas web viewer with WASD navigation on 1-2 captured rooms. POI click-to-place + text search. Documented capture SOP. Deck with architecture + accuracy metrics (PSNR/SSIM + tape-measure cm error, and the frame-selection-vs-uniform benchmark). *Proves the core.*
- **Round 2 (30 Nov 2026):** Full store-scale capture (tiled, LoD/Streamed SOG); mesh + recast-navigation-js navmesh + obstacle-aware pathfinding with clearance; shopping-list TSP routing; RoomFormer floorplan minimap; semantic search (LangSplat or Grounding-DINO product detection). Reloc recall benchmarked on 7-Scenes + own capture. Gamified capture UX.
- **Round 3 (on-campus 16-17 Dec 2026):** Same-day capture-to-deploy: warm 4090, tuned pipeline (target <2 h). Live web twin + navigation demo; sparse AR nav (ACE relocalization + VIO) demo in one aisle; measured accuracy live. Fallbacks rehearsed.

Recommendations

1. **Immediately:** stand up the frame-selector on EchoVoid's ARCore stack and get one room end-to-end (π^3 → gsplat 3DGS → SOG → PlayCanvas). Benchmark against uniform sampling to quantify the frame-selection gain — a graded, citable result.
2. **Inject ARCore poses into GLOMAP early;** verify metric scale against tape measure. De-risks scale/drift before it compounds.
3. **Build the mesh + recast-navigation-js navmesh track in parallel** with the splat viewer — shared reconstruction, different needs (photorealism vs collision/pathfinding).
4. **Scope AR nav to the cheapest credible version** (ACE reloc at decision points + VIO). Wire up EchoVoid's VisualMap.kt as the relocalizer. Continuous AR nav = one-aisle stretch demo.
5. **Ship the store-owner POI authoring tool** (click-to-place) as the primary POI mechanism; add open-vocabulary semantic search as a differentiator only after the pragmatic path works.
6. **Nail the capture SOP and gamified UX** — explicitly graded and aligned with the team's full-stack strengths.

Thresholds that change the plan: if reloc recall at (0.5 m, 5°) < ~70% in a real store → drop continuous AR nav, ship arrows-at-decision-points only. If a 2000 m² scene won't render at >30 FPS on a mid-range phone → force per-room tiling and desktop-first navigation. If the VGGT/ π^3 non-commercial *weight* license becomes an issue (unlikely for a hackathon) → fall back to MAST3R-SfM/GLOMAP-only init with the Apache-2.0 gsplat optimizer.

Caveats

- **VGGT and π^3 model weights are non-commercial research/education only** — fine for Techfest, but flag for any future commercialization; the gsplat optimizer (Apache-2.0) is unrestricted.
- Several performance numbers (3DGS optimization time, per-scene cost) are extrapolated from published per-scene figures and current GPU pricing; validate on your own captures.
- Consumer-app capabilities (Scaniverse/Polycam Android splats) shift frequently; re-verify before relying on any single app.
- ARCore Cloud Anchors has no official numeric accuracy; the ~0.16 m figure is a third-party proxy — measure your own reloc recall.

- Indoor VPS (ARCore Geospatial/Streetscape Geometry) is outdoor-only; do not depend on it inside a store.
- Wi-Fi RTT sub-meter results assume 802.11mc-capable APs, which a given store may not have — treat Wi-Fi as a coarse fallback, not a primary positioning source.
- The ARCore VIO drift figure (~0.5 m over 19.1 m) comes from a non-peer-reviewed blog on old hardware; the peer-reviewed *Sensors* 2022 benchmark gives ARKit 0.02 m/s but no clean ARCore %-of-length number — measure ARCore drift on your target device.

Cited papers, repos and tools

Name	Type	Link
VGGT (CVPR 2025)	Feed-forward 3D foundation model	https://github.com/facebookresearch/vgggt ; https://vgg-t.github.io/
π^3 /Pi3 (ICLR 2026)	Permutation-equivariant geometry	https://github.com/yyfz/Pi3 ; https://arxiv.org/abs/2507.13347
COLMAP	Incremental SfM/MVS	https://colmap.github.io/
GLOMAP (ECCV 2024)	Global SfM	https://github.com/colmap/glomap ; https://arxiv.org/abs/2407.20219
MP-SfM	Monocular-prior SfM (indoor low-parallax)	https://arxiv.org/abs/2504.20040
InstantSplat	Pose-free sparse-view 3DGS	https://github.com/NVlabs/InstantSplat ; https://arxiv.org/abs/2403.20309
3DGS (Kerbl 2023)	Gaussian Splatting	https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/
Hierarchical 3DGS (Kerbl 2024)	LoD large-scene splatting	https://repo-sam.inria.fr/fungraph/hierarchical-3d-gaussians/
2DGS (Huang 2024)	Meshable surfels	https://surfsplatting.github.io/
gsplat / nerfstudio	Apache-2.0 splat/NeRF framework	https://github.com/nerfstudio-project/gsplat
IVS3D	Intelligent video sampling for 3D	https://arxiv.org/abs/2110.11810
Adaptive Keyframe Selection (2025)	Learned keyframe selection	https://arxiv.org/abs/2510.23928
PlayCanvas SuperSplat + SOG	Web splat editor/format (MIT)	https://superspl.at/editor ; https://blog.playcanvas.com/playcanvas-open-sources-sog-format-for-gaussian-splatting/
splat-transform	Splat conversion/LOD/collision CLI	https://github.com/playcanvas/splat-transform
recast-navigation-js	WASM navmesh/pathfinding	https://github.com/isaac-mason/recast-navigation-js
three-pathfinding	A* over navmesh	https://github.com/donmccurdy/three-pathfinding
RoomFormer (CVPR 2023)	Floorplan reconstruction	https://github.com/ywyue/RoomFormer ; https://arxiv.org/abs/2211.15658
LangSplat (CVPR 2024)	3D language field on Gaussians	https://langsplat.github.io/ ; https://huggingface.co/papers/2312.16084
ConceptGraphs (ICRA 2024)	Open-vocab 3D scene graphs	https://arxiv.org/abs/2309.16650

Name	Type	Link
HOV-SG	Hierarchical open-vocab scene graphs	https://arxiv.org/abs/2403.17846
ACE (CVPR 2023)	Scene-coordinate regression reloc	https://github.com/nianticlabs/ace ; https://arxiv.org/abs/2305.14059
ACE-G	SCR generalization	https://arxiv.org/abs/2510.11605
hloc / SuperPoint+SuperGlue+NetVLAD	Feature-matching localization	https://github.com/cvg/Hierarchical-Localization
LaMAR (ECCV 2022)	Large-scale AR reloc benchmark	https://lamar.ethz.ch/ ; https://arxiv.org/abs/2210.10770
ARCore Geospatial / Streetscape / Depth	Google AR docs	https://developers.google.com/ar/develop/geospatial
ARCore Cloud Anchors	Persistent anchors	https://developers.google.com/ar/develop/cloud-anchors
Android Wi-Fi RTT (FTM)	Indoor ranging API	https://developer.android.com/develop/connectivity/wifi/wifi-rtt
WiNar (Wi-Fi RTT sub-meter)	802.11mc positioning study	https://www.sciencedirect.com/science/article/pii/S1574119221000687
Matterport Cortex AI	Commercial digital twin	https://matterport.com/cortex-ai
Pix4D capture best practices	Photogrammetry SOP	https://support.pix4d.com/hc/best-practices-for-image-acquisition-and-photogrammetry
Polycam interior capture guide	Capture SOP	https://poly.cam/blog/high-resolution-interior-photogrammetry-full-walkthrough-polycam-tutorial
Scaniverse (Android splats)	Consumer app	https://scaniverse.com/
RunPod / Vast.ai / Spheron GPU pricing	Cloud GPU cost	https://www.runpod.io/product/cloud-gpus ; https://getdeploying.com/gpus/nvidia-rtx-4090
Structured3D / 7-Scenes / Replica / ScanNet++ / ARKitScenes / Aria / Hilti	Eval datasets	(standard benchmark pages)